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Randomized Trial Comparing Pulmonary Vein Isolation Using the SmartTouch Catheter With or Without Real-Time Contact Force Data

Ullah W[1], McLean A[1], Tayebjee M[2], Gupta D[3], Ginks MR[4], Haywood GA[5], O'Neill M[6], Lambiase PD[7], Earley M[1], Schilling R[1], on behalf of the UK Multicenter Trials Group[8].

[1]Cardiology Research Department, St Bartholomew's Hospital, Barts Health NHS Trust, London, UK.

[2]Leeds General Infirmary, Leeds.

[3]Liverpool Heart and Chest Hospital, Liverpool.

[4]John Radcliffe Hospital, Oxford.

[5]Plymouth Hospitals NHS Trust, Plymouth.

[6]Cardiovascular Division, Guy's and St. Thomas' NHS Foundation Trust, London.

[7]The Heart Hospital, London.

[8]See Appendix for Members of the UK Multicentre Trials Group

Corresponding Author:

Waqas Ullah

Cardiology Research Department

St Bartholomew's Hospital

London EC1A 7BE, UK

Tel: +442034656681

Fax: +442034655771

E-mail: waqas@doctors.org.uk

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Abstract

Background: Contact force (CF) information may improve the safety and efficacy of ablation for paroxysmal atrial fibrillation (PAF).

Objective: Assess the impact of CF data on ablation for PAF.

Methods: Patients undergoing first-time PAF ablation were randomized at 7 UK centers to ablation with (CF-on) or without (CF-off) CF data available to the operator, using the same ablation catheter and mapping system. An ablation CF of 5-40g was targeted. Pulmonary vein (PV) reconnection was assessed with adenosine at 60 minutes. Follow-up for arrhythmia recurrence was for one year with 7-day Holter recordings at 6 and 12 months.

Results: 117 patients were studied (59 CF-on, 58 CF-off). In the CF-on group, a reduction in acute PV reconnection rates (22% versus 32%, P=0.03), but no significant difference in 1 year success rates off antiarrhythmic drugs (49% versus 52%, P=0.9) was observed. There was no difference in major complication rates: 2/59 (3%) CF-on, 3/58 (5%) CF-off, P=0.7. Procedure and fluoroscopy times were not significantly different (P>0.5). Overall mean CFs per ablation were not different between groups (13.4[9.1-19.6]g CF-on, 13.4[7.4-22.4]g CF-off, P=0.5), but a greater proportion of readings in the CF-on group were in the target range (80% versus 68%, P<0.001).

Conclusion: This randomized, multicenter study demonstrated that CF data availability was associated with reduced acute PV reconnection but not improved 1 year success rates, procedure and fluoroscopy times, or complication rates. There was a reduction in extremes of CF, above and below the study target range, suggesting greater CF control during ablation.

Key Words: Atrial Fibrillation; Catheter Ablation; Contact Force Sensing; Randomized Controlled Trial

Clinical Trials Identifier: NCT01730924; http://clinicaltrials.gov/.

Introduction

Increasing catheter contact force (CF) during radiofrequency ablation for atrial fibrillation (AF), while increasing ablation efficacy[1–3], is associated with an increasing incidence of steam pops and perforation[1,4]. Conversely, lower CF during ablation is predictive of pulmonary vein reconnection sites[5–7]. During mapping and ablation of AF, a high degree of variability in CFs is observed between different operators and at different atrial sites[8,9]. Therefore, access to CF data may improve the safety and efficacy of clinical ablation procedures. With the use of CF-sensing catheters for ablation, success rates are improved in some reports[10,11], but not others[12–16], along with reductions in procedure[10,12–15,17] and fluoroscopy times[10,13,14,16–18]. However, only one of these studies, a small single site 38 patient study[12], was randomized using the same equipment (mapping system and ablation catheter) in both arms.

We sought to measure the impact of CF-sensing on the AF ablation procedure in a prospective, randomized, multicenter study. All patients had paroxysmal atrial fibrillation (PAF) and in all cases the Carto3 electroanatomic navigation system was used with the Thermocool SmartTouch CF-sensing catheter (Biosense Webster Inc., Diamond Bar, CA, USA). The primary hypothesis tested was that availability of CF data would reduce the time to perform pulmonary vein (PV) isolation. Secondary end points were fluoroscopy times, procedure times, CF variability during ablation, acute success rates (judged by PV reconnection), complication rates and one year success rates off antiarrhythmic drugs (AADs).

Methods

The study was performed at 7 different UK centers. All patients gave informed consent to participate, which was approved by the UK National Research Ethics Service and reported according to CONSORT guidelines[19]. Subjects could be included if over 18 years old with symptomatic paroxysmal AF (PAF) as defined by current guidelines[20]. Exclusions included previous ablation for AF, contraindications to anticoagulation, severe left ventricular systolic dysfunction (ejection fraction <35%), prior mitral valve replacement or hypertrophic or infiltrative cardiomyopathy. Participants were randomized using computer-generated block randomization.

Procedures were performed in the post-absorptive state under intravenous moderate conscious sedation or general anesthesia and undertaken by physicians who had performed at least 50 point-by-point AF ablation procedures as first operator. Uninterrupted oral anticoagulation was used in all cases. Trans-septal punctures were performed using an Endry's co-axial (Cook Medical, IN, USA) or Brockenbrough needle (St. Jude Medical, MN, USA). A Lasso Nav circumferential steerable pulmonary vein (PV) mapping catheter (Biosense Webster Inc.) and the SmartTouch catheter were passed into the left atrium. Steerable sheaths were discouraged and only permitted if the operator felt they could not complete the case without one. Intravenous heparin was used to maintain an activated clotting time of 300-400 seconds. Ablation was performed in temperature-controlled mode (temperature limited to 48˚C, power to 30W). The irrigation flow rate of the SmartTouch catheter was restricted to 17 ml/min during ablation. Radiofrequency energy was delivered for at least 20 seconds at a location with the aim of each ablation being a reduction in local bipolar electrogram amplitude by >80% or until <0.1 mV[21].

In the CF-on group, CF data were visible to the operator, while in the CF-off group the operator used a SmartTouch catheter but was blinded to the contact force. This ensured that the catheter handling characteristics were the same for both patient groups. In the former group, a CF range during ablation of 5-40g was targeted. As excessive CF is associated with a risk of myocardial perforation[22], in both groups, a visual warning alerted the operator if the CF exceeded 50g.

In both groups, antral PV isolation (PVI) was performed by wide area circumferential ablation (WACA). The endpoint for WACA ablation was entry and exit block, assessed using the PV mapping catheter. A 60 minute waiting time was used to assess for PV reconnection. If at this time, the veins remained isolated, each vein was individually assessed for reconnection, using a circular mapping catheter, by the intravenous administration of 18mg adenosine. In the presence of reconnection, further ablation was performed to re-isolate the vein. Those veins where the waiting time was not fulfilled (unless spontaneous reconnection occurred before 60 minutes had elapsed) or where adenosine was not administered were regarded as protocol violations for the acute reconnection analysis and excluded from that analysis.

In patients with documented atrial flutter, ablation of the cavo-tricuspid isthmus was also performed. No other linear lesions or ablation of complex fractionated electrograms was undertaken, and no attempt made to look for extra-PV triggers.

A post-procedural transthoracic echocardiogram was performed to rule out pericardial effusion.

Follow up

Anti-arrhythmic drugs (AADs), including amiodarone, were stopped at discharge and all patients followed up at 3, 6 and 12 months. Post-procedure, a 3 month blanking period was used. Patients with a recurrence of symptoms and/or documented atrial tachyarrhythmia after this period were offered a repeat procedure. 7-day Holter monitors were performed at 6 and 12 months. Complications were recorded from the time of the procedure until discharge and at follow up visits, and reported according to guidelines[20]. Clinical success was defined as freedom from symptomatic or asymptomatic AF or atrial tachycardia lasting ≥ 30 seconds off AADs[20].

Contact Force Analysis

The CF data were exported from Carto3 and analyzed using Matlab (MathWorks, Natick, MA, USA). CF data were recorded at 20Hz during ablation. Only the CFs during the initial isolation of the PVs were included in the analysis.

End points

The primary study endpoint was the time taken to isolate both sets of veins. Secondary end points were fluoroscopy times, procedure times, acute success rates (judged by acute PV reconnection), complication rates (with complications as defined by HRS guidelines[20]) and clinical success rates.

Statistical analysis

Statistical analysis was performed using SPSS (IBM SPSS Statistics, Version 20 IBM Corp, Armonk, NY, USA) and Matlab V7.12 with Statistics Toolbox V7.5. A p-value <0.05 was considered statistically significant. The sample size was calculated for the primary end point (time taken for PV isolation). Assuming an α error of 0.05 and β error of 0.2, 60 patients were required in each group to detect a 25% reduction in time for PVI, assuming a mean of 112 minutes for PVI (without CF information) with a standard deviation of 54 minutes. These latter times were derived from retrospective analysis of the AF ablation database at St Bartholomew's Hospital. Patients were compared using an intention-to-treat analysis with respect to their CF group, but those randomized to a treatment group who had no procedure were excluded - as this was not felt to introduce bias as comparisons were procedure-based[23]. Groups were compared using parametric and non-parametric testing based on the distribution of the data. Results are presented as mean±standard deviation (SD) or median [interquartile range]. Categorical variables are described as count and percentage and compared using Chi-squared or Fisher's exact tests. A P value of <0.05 was considered significant.

Results

Patient progression through the trial is illustrated in Figure 1. One patient in the CF-on group did not have a procedure performed as part of the study as the SmartTouch catheter was subject to a temporary manufacturer recall on the day of their procedure. Two patients in the CF-off group did not have study procedures – due to technical problems with the Carto3 system and the SmartTouch catheter respectively. These three withdrawn patients are excluded from the analysis. Three patients, all from the CF-off group, did not complete 12 month follow up. All three patients are alive without complications based on telephone conversations, one has had a redo procedure at 6 months while the other two denied symptoms at one year. The baseline demographics of the study population are displayed in Table 1: there were no significant differences between the study groups.

All the operators in the study had used CF-sensing catheters prior to study commencement, most having performed over 30 cases, while all had performed at least 5.

The procedural results are presented in Table 2, and complications in Table 3. In only one case was a steerable sheath used – to complete the left WACA in a CF-off case. The availability of CF data did not result in any differences bilateral PVI times (the primary endpoint), nor did it significantly influence procedure or fluoroscopy times, radiation doses or complication rates.

Acute and Long Term Success

76/468 (16%) PVs were excluded from the acute reconnection analysis due to protocol violations in the 1 hour waiting time and/or the administration of adenosine: 37/236 (16%) CF-on, 39/232 (17%) CF-off, P=0.8. Of the PVs included, 104/392 (27%) reconnected acutely – 68/104 (65%) spontaneously by 60 minutes and 36/104 (35%) with adenosine. The availability of CF data was associated with a significant, 31%, relative reduction in the rate of acute reconnection (Figure 2).

114/117 patients completed 12 months of follow up. In the CF-on group, the 12 month single procedure success rate off AADs was 29/59 (49%). For the CF-off group, this was 29/56 (52% - including the patient who did not complete follow-up but was known to have a redo procedure, but excluding the other 2 that did not complete follow-up), or 29/58 (50%, if all those that did not complete follow-up were counted as treatment failures). A Kaplan-Meier curve comparing the groups is presented in Figure 3.

Contact Force Analysis

CF data were available for 81/117 cases (40 CF-on; 41 CF-off): in the remainder of studies the Carto study backups were corrupted at the source centers and so could not be analyzed.

If all of the individual CF measurements (4,048,039) made during the WACA to initially isolate the veins were compared, differences were evident in the spread of CFs between groups (Figure 4). In the CF-on group, the CF during ablation was maintained within the 5-40g CF threshold specified in the protocol for a greater proportion of the time (80% compared with 68% in the CF-off group, $P<0.0005$).

There were 2,587 individual radiofrequency applications included in the CF analysis. Comparing the two groups, while there was no significant difference in the mean CF for a radiofrequency application, each application was longer and reached a higher FTI in the CF-on group (Table 4).

Discussion

In this randomized controlled multi-center trial, the availability of CF data to the operator during an ablation procedure for PAF was not associated with any significant reduction in procedure or fluoroscopy times. There was a significant reduction in acute PV reconnection at 1 hour with adenosine but no difference in 1 year success rates. There was also a significant difference in the spread of CF during ablation, with less CF measurements outside the 5-40g target (20% versus 32%, $P<0.001$).

Previous studies have demonstrated differences in procedural parameters between different mapping systems[24] and different versions of the same mapping system[25]. In the current study, the same electroanatomic navigation system, and importantly the same version of this, was used in both groups. Furthermore, the actual catheter used and the ablation power and irrigation settings were kept the same in each group. Where different ablation catheters have been used in studies in the two comparison groups[10,11,13,14,16–18], it may be that some of the differences observed are secondary to the performance of the CF-sensing catheter *per se* rather than the CF data it records. Lastly, the study population included was only PAF patients, also serving to make the comparison purer with respect to the impact of the addition of CF data. Previous small studies[5,12,15] have investigated patients using the same procedural equipment in CF blinded and unblinded groups (only one of which was randomized[12]).

The availability of CF data did not improve long term success rates in AF ablation. This is consistent with previous work studying manual AF ablation in some studies[12–16] but not others[10,11]. There was an improvement in acute reconnection rates and it may be that some of the long term negative impact of suboptimal ablation in the CF-off group was counterbalanced by corrective further ablation for reconnection acutely after a 60 minute wait and adenosine. The benefits of CF data may be more apparent in the long term in a clinical setting where less stringent investigation for acute reconnection is conducted.

Work published subsequent to the inception of the current trial has sought to establish optimal CF targets for ablation, based on pulmonary vein segmental reconnection[5,6,9,26] or impedance drop during individual ablations[2,3]. In the current study, only a broad CF range was specified for operators to target. The use of more refined targets, relating not just to the amount of contact[5,16,27] but also its duration (as incorporated in the FTI[2,6,9]) and quality (including stability and continuity[3,26]), as well as ablation power (as incorporated in the Ablation Index[28]) may have resulted in a significant improvement in success rates. The attainment of such targets could be facilitated by the use of automated lesion marking software[29], which only places lesion markers when certain user-defined criteria are attained.

In the current study, no difference was noted in the mean CF applied between the two groups. This contrasts with previous work where the mean CF was lower when the operator was blinded to CF data[5,12]. One possibility for the smaller difference is that experience with the use of

CF-sensing catheters trains operators to change their approach to ablation even when CF data is not available. The operators in the current study had more prior experience with CF sensing catheters than those participating in earlier studies and there is some evidence that accumulating experience with CF-guided ablation improves CFs even in CF-blinded procedures[12]. While there was no overall difference in the mean CF for each ablation, there were more measured CFs during ablation within the target CF threshold of 5-40g in the CF-on group. The absence of CF data had a greater impact on CF below the target range: in the CF-off group 25% of the measured CFs were <5g, compared with 17% in the CF-on group. Such low forces would one expect be associated with ineffective ablation. This effect of the presence of CF data was more marked than for CFs >40g (6% CF-off versus 4% CF-on), perhaps partly due to the high force (>50g CF) visual safety indicator used in both groups. These results suggest that availability of CF data results in more control over the CF during ablation, by minimizing extremes of CF, consistent with previous work where a different range was targeted[15]. The mean CF values would also be less sensitive to picking up differences in the spread of the CF from the availability of CF data.

Control over CF during ablation is important from the viewpoint of safety as higher CF is associated with steam pops during ablation and perforation[1,4]. In the current study though, there was no significant difference between the two groups in the complication rates, despite the greater proportion of CF readings in the CF-off group >40g. No difference in complications rates is consistent with previous retrospective studies[17,18]. One possible reason for the lack of difference in the current study is that the main major complication CF data availability would conceivably reduce is tamponade (rather than, for example, vascular access complications). The current study is underpowered to examine a difference in such a relatively low incidence complication.

The lack of effect of the availability of CF data on procedural parameters is surprising when considering previously published data showing that the use of real time CF data was associated with improvements in procedure[10,12–15,17] and fluoroscopy times[10,13,14,16–18], (though one group has demonstrated the converse[11]). It might be expected that availability of CF data would shorten procedures as less poor contact, ineffective ablations would be delivered and so PV isolation could be achieved faster. A reduction in fluoroscopy times was hypothesized to occur as the operator would be less reliant on fluoroscopy to judge catheter contact. In the current study, the procedure and fluoroscopy times were numerically lower in the CF-on group but not significantly. The latter is despite the additional reablation for the increased number of acute reconnections in the CF-off group. Interestingly, the ablation time was longer in the CF-on group, both overall and for each individual ablation. This may reflect greater confidence on the part of the operator in ablating when CF data is available. It is important though that maximal ablation targets[2,6] are not exceeded during ablation for safety reasons and the availability of CF data would help to ensure this when these targets are used.

Limitations

The presence of the visual alarm in both groups for CF considered dangerously excessive meant that the CF-off group was not fully blinded to CF. Having this alarm, and thereby not compromising the safety in the CF-off group, while a scientific compromise, was accepted on an ethical basis. The operators in the study all had prior experience with CF-sensing catheters: the results may therefore not be representative of those for operators naïve to the technology. In the CF-on group, while CF was in the target range for a greater proportion of the time (80%) than the CF-off group, it was not in range continuously (likely for practical reasons such as LA anatomy or patient respiration). Previous work has suggested that maintaining a CF within target range ≥80% of the time improves outcomes[30] and therefore the percentage in this trial is reasonable. Wide CF targets were used in the current study and ablation was guided by electrogram attenuation (as used in previous work previously[21,31]). Work published subsequent to the conception and commencement of the trial has suggested FTI may be a better target[6] rather than electrogram attenuation[3].

Conclusion

In this randomized study of PAF patients undergoing their index ablation, the availability of CF data did not offer any advantages in terms of procedure, fluoroscopy times or complication rates, though the incidence of CFs during ablation outside the target range (especially below this range) was reduced. This suggests that the availability of CF data, while it improves catheter control, does not impact on clinical outcome, likely, because it is only one of multiple factors that determine lesion efficacy and safety. The RF application power and time, as well as the tissue characteristics are also important variables. Technologies that help the operator visualize all of these during a procedure are likely to be required for CF data to really have an impact on clinical outcome.

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Tables

Table 1

Baseline Demographics

Parameter	CF-off	CF-on	P-value
Patients	58	59	
Age (years)	59±11	59±10	0.6
Male Gender	34/58 (59%)	33/59 (56%)	0.9
AF Duration from diagnosis (months)	27 [16-65]	42 [18-75]	0.3
European Heart Rhythm Association AF Symptom Score	2.6±0.5	2.7±0.7	0.6
Left Atrial Diameter (mm)	38±5	41±5	0.07
Number of current and previous antiarrhythmic drugs	1 [1-2]	1 [1-2]	0.5
Previous or current amiodarone use	12/58 (21%)	9/59 (15%)	0.5
Hypertension	22/58 (38%)	14/59 (24%)	0.1
Ischemic Heart Disease	2/58 (3%)	2/59 (3%)	1
Cerebrovascular Disease	5/58 (9%)	2/59 (3%)	0.3
Diabetes	5/58 (9%)	1/59 (2%)	0.1
CHA_2DS_2VASC Score	1.8±1.5	1.3±1.5	0.1

Table 2

Procedural Factors

Factor	CF-off Group	CF-on Group	P-value
Number	58	59	
Use of General Anaesthesia	18 (31%)	18 (31%)	1
CTI line	7 (12%)	8 (14%)	1
Total Procedure Time (mins) *	196 [165-217]	194 [171-220]	0.96
Total PVI Time (mins)*	73 [4-86]	70 [55-90]	0.82
Left WACA Time (mins)	33 [25-47]	37 [27-46]	0.57
Right WACA Time (mins)	33 [24-45]	32 [24-50]	0.92
Total Fluoroscopy Time (s)*	830 [400-1415]	648 [354-1867]	0.82
Total Fluoroscopy Dose (cGy.cm^2) *	894 [302-1731]	789 [345-2235]	1
Total Radiofrequency Ablation Time (s) *	2315 [1890-2782]	2483 [2055-2973]	0.86

*In the case of 1 CF-off and 1 CF-on case, the procedures were terminated early: in the former case due to a fault in the ablation generator and in the latter due to the patient developing cardiac tamponade. In both cases, procedures were terminated after completion of the left WACA. Therefore for these patients, their curtailed data are not included in the whole procedure data.

PVI=Pulmonary Vein Isolation; WACA=Wide Area Circumferential Ablation

Table 3

Complications per group with severity assigned as per guidelines[20]

Complications	CF-off	CF-on	P-value
Major	3/58 (5%)	2/59 (3%)	0.7
	1 Hematoma requiring readmission 1 Pericarditis with a small effusion prolonging hospitalization by > 48 hours 1 Broken femoral venous access sheath requiring removal by interventional radiology	1 Tamponade 1 Pseudoaneurysm	
Minor	2/58 (3%)	4/59 (7%)	0.7
	2 Hematoma	1 Pericardial effusion 1 Hematoma 2 Pericarditis	

Table 4

Individual radiofrequency application parameters

Parameters for individual radiofrequency applications	CF-off	CF-on	P-value
Mean Ablation CF (g)	13.4 [7.4-22.4]	13.4 [9.1-19.6]	0.5
Ablation Duration (s)	43.2 [29.9-76.0]	52.3 [31.1-102.8]	<0.0005
Ablation FTI (g.s)	639 [303-1301]	750 [372-1575]	<0.0005
LWACA Mean Ablation CF (g)	12.6 [7.1-21.1]	12.2 [8.3-17.6]	0.45
LWACA Ablation Duration (s)	41.0 [29.9-71.6]	50.0[29.8-103.0]	<0.0005
LWACA Ablation FTI (g.s)	571 [282.1-1182]	705.4 [317-1460]	0.007
RWACA Mean Ablation CF (g)	14.9 [8.5-23.5]	14.9 [10.0-20.8]	0.9
RWACA Ablation Duration (s)	45.4 [29.6-77.9]	52.9 [32.9-102.3]	<0.0005
RWACA ablation FTI (g.s)	723 [349-1478]	814 [418-1709]	0.005

CF=Contact Force; FTI=Force Time Integral; LWACA=Left Wide Area Circumferential Ablation;

RWACA=Right Wide Area Circumferential Ablation

Figure Legends

Figure 1

Progress of patients through the trial

Figure 2

Acute reconnection (as determined by per protocol assessment) by CF group for all of the pulmonary veins, the left and right sided veins. *P=0.03.

Figure 3

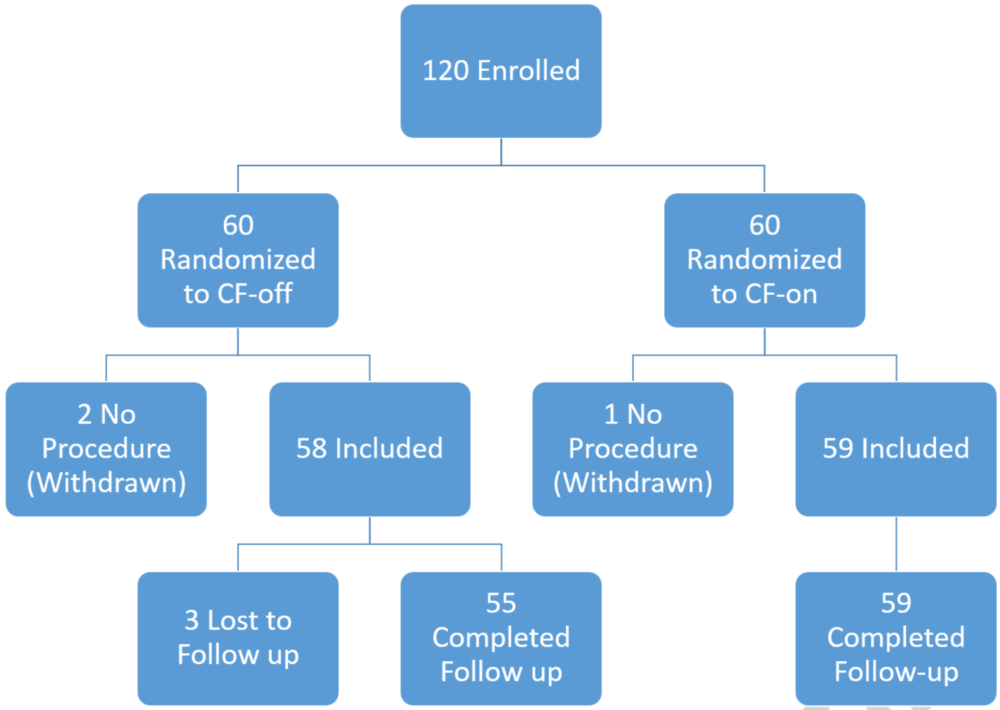
Kaplan-Meier survival curves comparing long term outcomes in both groups. One patient in the CF-off group did not complete follow-up but was known to have had a redo-procedure at 6 months and therefore was taken as a treatment fail at 6 months. Two further patients in the same group were censored from when they failed to attend follow-up.

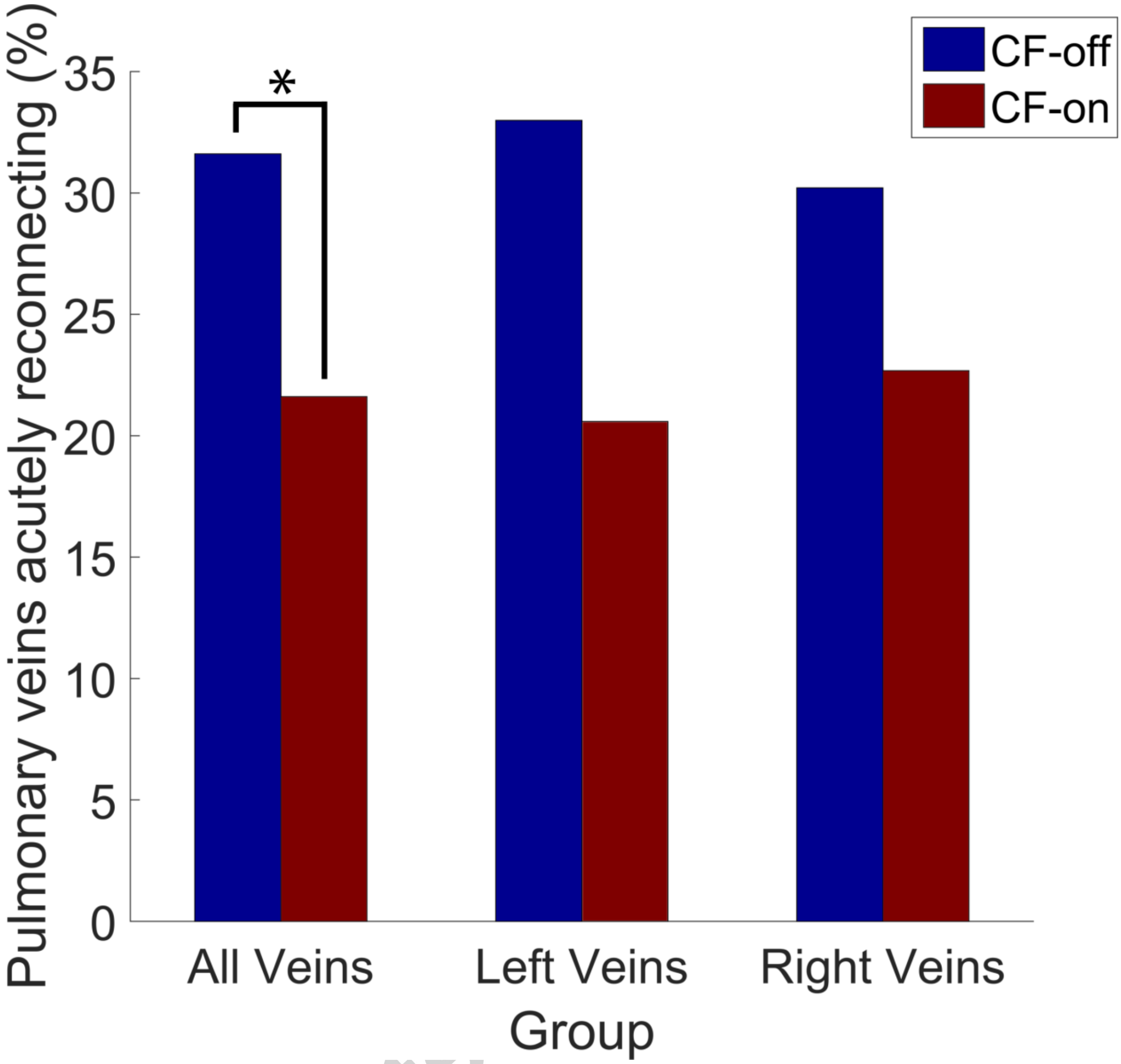
AF=Atrial Fibrillation, AT=Atrial Tachycardia, AAD=Antiarrhythmic Drugs

Figure 4

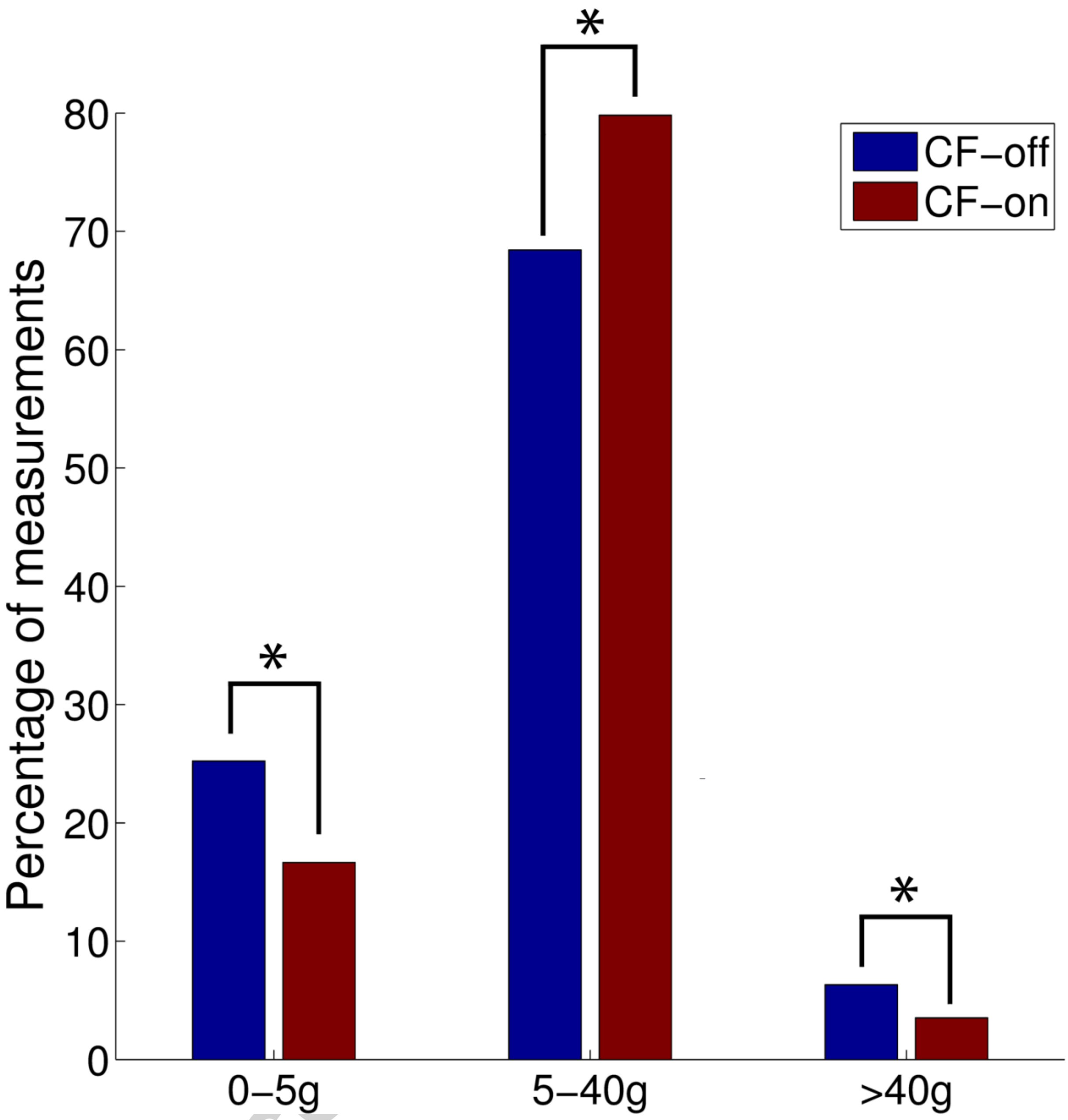
Contact force (CF) measurements during ablation: 4,048,039 CF measurements are presented in the bar chart. The CF measurements are subdivided according to whether they were below, within or above the study target CF range

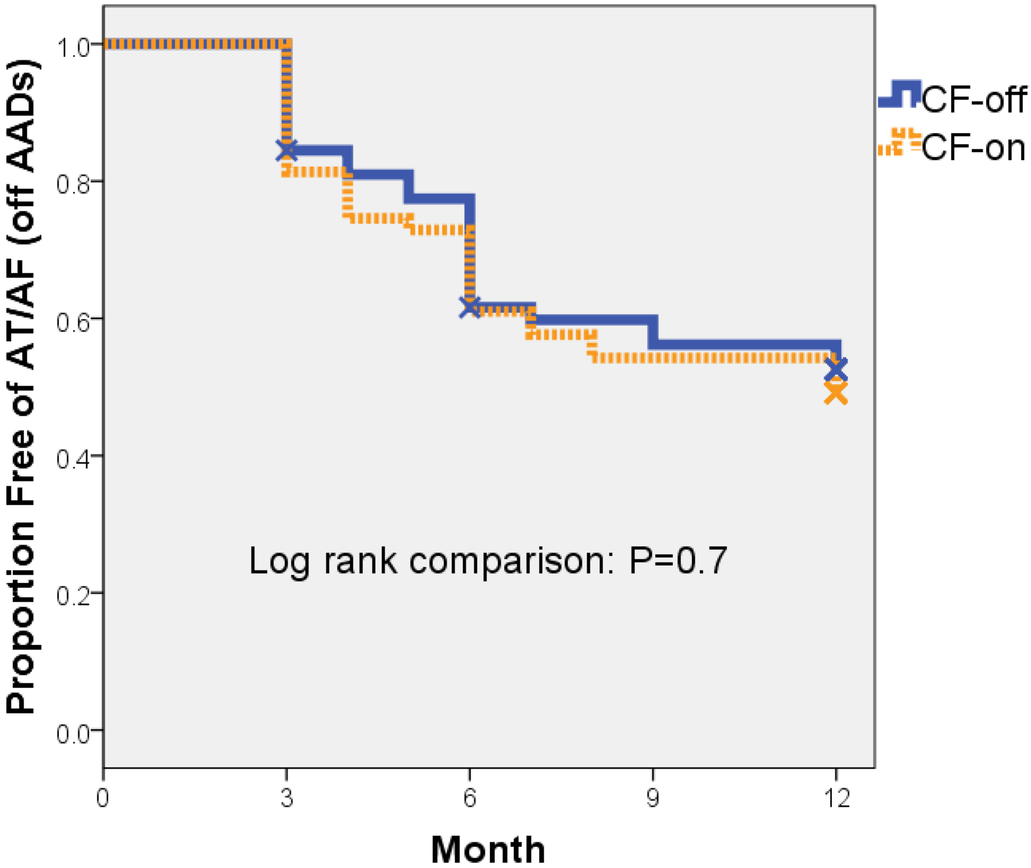
*=P<0.0005 for comparison of the CF measurements in the two groups.


120 Enrolled
60 Randomized to CF-off
60 Randomized to CF-on
2 No Procedure (Withdrawn)
58 Included
1 No Procedure (Withdrawn)
59 Included
3 Lost to Follow up
55 Completed Follow up
59 Completed Follow-up



Pulmonary veins acutely reconnecting (%)
35
30
25
20
15
10
5
0
*
CF-off
CF-on
All Veins
Left Veins
Right Veins
Group


Percentage of measurements
0
10
20
30
40
50
60
70
80
0–5g
5–40g
>40g
CF–off
CF–on
*
*
*



Proportion Free of AT/AF (off AADs)
1.0
0.8
0.6
0.4
0.2
0.0
0
3
6
9
12
Month
CF-off
CF-on
Log rank comparison: P=0.7